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(1) Research Program/Project/Study Title: Innovation Ecosystem Assessment in SUCs: A Case of Capiz State University	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	ROXAS CITY
Province:	CAPIZ
Region:	VI

(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☒ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity	(8) Discipline/Field of Specialization
	Social Science
(8A) Abstract This study examines Capiz State University's (CAPSU) Innovation Ecosystem, focusing on faculty research capabilities. It aims to identify faculty research capabilities, assess stakeholder participation, conduct a descriptive analysis of demographics and research parameters, formulate an enhanced training program, identify relevant technology/software, and explore challenges in research output creation. Anchored in Tolentino's self-assessment scale and Kotesmir & Shashnov's research quality criteria, the Triangulation Design integrates quantitative surveys, qualitative interviews, and primary data analysis. Findings unveil commendable research engagement at CAPSU, tempered by challenges like time constraints and skill gaps. Recommendations propose tailored training programs, strengthened financial support, collaborative structures, and technology training for enhanced research proficiency. This holistic approach, fostering an environment conducive to impactful research, contributes to the theoretical body of knowledge by addressing challenges faced by faculty researchers in a Philippine State University, providing insights for research capabilities, and advancing the nation's innovation goals. This study uniquely contributes by delving into the nuanced dynamics of CAPSU's innovation ecosystem, providing a detailed understanding of faculty research capabilities, and proposing targeted strategies for enhancement.	
(8B) Keywords: <i>Innovation Ecosystem, Faculty Research, Triangulation Design, Research Proficiency</i>	

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(9) Rationale

The Philippines is currently in the 51st place in the 2021 Global Innovation Index rankings and is considered a Lower Middle-Income Group country. Based on its accelerated growth of being from the 91st to the 51st rank, the Philippines is considered to be an emerging economy through its efforts in shaping its innovation landscape. Currently, the Philippines is at the top quartile in terms of Knowledge, and Technology, and Business Sophistication but is ranked lower when it comes to its Institutions, Human Capital and Research, Infrastructure, Market Sophistication, and Creative outputs (Dutta et al., 2021). Nevertheless, institutions such as Capiz State University have been addressing and taking action regarding the gaps in order to enhance the different pillars that build up the country's innovation index.

Producing quality research and technology has always been an integral part of the long-term goals of CAPSU. Abundant raw resources such as various seafood produce, crops, native poultry, and livestock are at the disposal of CAPSU's faculty researchers. Thus, with the diversity of readily available resources, it is up to the faculty researchers to innovate around these products and create a production chain that will boost the quality of life, exporting power and livelihood of the Capizeños. Based on the data provided by the Philippine National Statistics Office (n.d.), over 727 farming plots are not being utilized, and 61,219 are used for temporary farming. This could be an opportunity for the researchers of CAPSU to develop viable technologies that could utilize the full potential of unused lands through research-based projects and development.

With that said, in order to be able to provide quality services and results to the community, CAPSU's faculty researchers must be able to produce quality research and possible results that will have a significant impact on the scientific community. Bowsher et al. (2019) highlighted the three main narratives that emerged during their study on developing the health research capacity. These themes were the "effect of power relations on capacity development, stronger links between research, policy and practice, and the importance of systems approach." Although this system of findings may vary from institution to institution, this can be used as a component of determining the research capacity of the institution's faculty researchers.

At the start of every year, the CAPSU Research, Development, and Extension (RDE) Training Office are regularly conducting a Training Needs Assessment (TNA) for its faculty researchers, which assesses the training and capability-building activities that they need in order to enhance their capacity in producing quality research. In hindsight, it is hypothesized that the faculty researchers have a level of awareness of their own capability from conceptualization up to the publication process of their outputs. Nevertheless, research findings regarding the research capability of CAPSU's faculty researchers are still nonexistent. Therefore, the RDE office made this their priority in order to help the faculty researchers achieve their roadmap goal and contribute to the success of their Innovation Ecosystem.

(10) Objectives/Statement of the Problem

This study aims to assess the Innovation Ecosystem of Capiz State University. Specifically, it seeks to achieve the following objectives.

1. Identify the level of the research capability of the CAPSU Faculty.
2. Identify the level of the extent of participation of other stakeholders.
3. Descriptive analysis of demographics and other pertinent papers.
4. Formulate an enhanced training program that can address the needs of the faculty researchers in conducting their studies.

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5. Identify the possible technology/software that can help the faculty researchers in their research efforts.

6. Determine the problems/issues encountered in making and conducting a research output.

(11) Review of Literature/Prior Art Search

Parameters for Creating a Quality Research Paper

Specific parameters must be met in order to create a high-quality research output. As cited by Chu et al. (2014), there are four prerequisites to do so. First is the individual skills of each researcher, which boil down to their ability to write, analyze, synthesize, and utilize their readily available research tools. Next is the appropriate infrastructure, which includes but is not limited to: the work environment, research equipment, laboratory, and others. The third prerequisite is the research's relevance to a nation's short-term and long-term goals and policies. This is one of the main determining factors of whether or not a research proposal will be supported or funded by the researcher's institution or partner agencies. The final prerequisite is the perceived impact or its contribution to the global/international research efforts and needs. It was further stated that the African Researchers could barely reach international standards due to the lack of these prerequisites and support from different funding agencies. Due to these factors, capable researchers are being exploited by other peers, institutions, and agencies, using the "Publish or Perish" mandates of their government as well as the government.

Difficulties of Faculty Researchers in Conducting Research

In the Philippines, pursuing personal and professional growth is a challenge due to heavy workloads, multiple designations, lack of capacity-building activities, and lack of funding (especially for publications) due to the unrevised policies of the government agencies is the most difficult hurdle that a faculty researcher can face (Vecaldo et al., 2019). The "Publish or Perish" mandate also exists in the country as it requires the faculty of Higher Education Institutions (HEIs) to publish in ISI/Scopus-indexed journals and publishers in order to be promoted to higher levels of position classification and to receive compensation (DBM, n.d.).

This puts a lot of unnecessary pressure on the HEI faculty to publish in such journals using outdated methodologies, lack of research proficiency due to poor funding, endangering one's social identity, lack of research proficiency due to heavy workload, and other factors (Kurt, 2018). This is also evident within CAPSU because it is a State University and Colleges (SUCs). Therefore, it is subject to the government's rules, laws, and policies regarding career advancement. It was only until the year 2021 that CAPSU could hire personnel that could handle specific tasks of its faculty researchers that are integral to the success of their paper, such as external and internal publication.

Engaging Faculty Researchers to Conduct Research

According to Huenneke et al. (2017), established research institutions practice adding newly hired greenhorn faculty to existing studies and projects in order for them to learn more in their field and to create research outputs. In their study, the participants had a standard model on how to increase the research activity within their institution. These models are: "cluster hiring, revision of institutional policies and infrastructure, and systematic prioritization of research for administrators as well as for individual faculty members."

From the local perspective, the revision of policies is currently underway and is the primary objective of the RDE Office. The systematic prioritization had already commenced due to the 5-year road map and innovation ecosystem that was formulated by the entire RDE body.

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Based on Huenneke's findings, investing in institutional equipment, components, and resources or by means of acquiring external funding for institutional development can greatly contribute to its long-term goal than recruiting or hiring new faculty researchers from other institutions. These supported institutions should have a program focused on career development plans (aligned with research) for junior and new faculty members. This way, the universities would then have a pool of talented individuals that can greatly contribute to their future research efforts and their country's national agenda.

In the same study, Huenneke stated that Transition and Succession planning should be heavily considered. In relation, CAPSU also changes its assigned leaders annually. This can cause shifts of interest in research agendas, extension programs, and other aspects of the RDE Office. Transition and succession had been a common practice within the institution (through verbal or internal agreement), but no formal policies or advisories exist. This could also be a determining factor to the research capacity of the faculty researchers.

(12) Scientific Basis/Theoretical Framework

This study is anchored on the study of Tolentino (2021) where he utilized the perceived level of the research capability of their science teachers in basic education using a self-assessment scale adopted from Abrazo & Coronel (2016). The results of their study provided them the insight of what are the existing gaps and concerns they should address in order to enhance the capability of their faculty researchers.

Although this is only a self-perceived scale, the researchers of this study have also taken into consideration the external perspective in perceiving the research capacity of a researcher in order to have empirical evidence used for additional policy enhancement, capability building activities, program enhancement proposals, and other inputs for the development of the faculty researchers of CAPSU.

Therefore, the analysis of the empirical evidence of this study anchored on the study of Kotesmir & Shashnov (2017) to assess the quality of their research output. Within their study, they have analyzed the following aspects: Dynamic Publication Activity, Citation Indicators, the University's Integration into International Research Collaboration, Individual's Research Capacity, and the Journal Quality of their published papers. Additionally, components such as research quality, similarity index, and grammar quality is considered for analysis.

(13) Methodology

Research Design

This study employed qualitative and quantitative research with the Triangulation Design: Validating Quantitative Data Model Cresswell, 2014). Through this design, the faculty researchers' self-perceived research capacity (Quant) further explored through a semi-structured interview regarding their strengths and difficulties in conducting research (Qual). In turn, it further validated using their research output, citations, and other research-based activities in order to create pragmatic results of their capacity.

Research Instrument

This research utilized a survey questionnaire adapted from the study of Tolentino (2021) which includes the overall profile of the faculty researchers. The research instrument has undergone a validity and reliability test. This study also integrated the components of the CAPSU RDE Training Office's Training Needs Assessment, which expanded the functionality of the stated survey questionnaire.

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For the quantitative data, researcher survey questionnaires were utilized to cater to the strengths, weaknesses, and difficulties of the faculty researchers within the institution. Lastly, primary data consisting of their research outputs, proposals, publications, and other research-related outcomes are analyzed in order to contrast side by side with the results of the first two data sets.

Data Collection

Permission from the Campus Administrators, Satellite College Directors, and the RDE Office was requested prior to conducting the study. The selected participants were given a consent form along with the attached survey questionnaire. In turn, the results of the survey used to deduce the necessary questions for the qualitative interview.

After validating both quantitative and qualitative results, a letter of request were sent to the RDE office to allow the researchers to procure the primary data from the faculty researchers for further analysis.

Respondents of the Study

This research will utilize Multi-Stage Sampling based on the standard set by the RDE Office, for which they consider a faculty researcher of CAPSU, which narrows it down to a set of key informants and participants. This is the most appropriate sampling technique to attain the objectives of this study. In turn, this improved the trustworthiness and timeliness of the study.

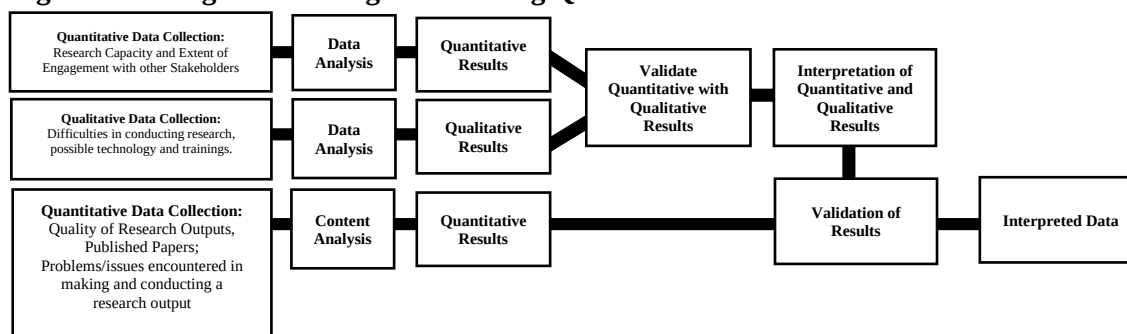
Table 1. Inclusion and Exclusion Criteria

Inclusion Criteria	Exclusion Criteria
The respondent must be a regular faculty that is required to do research.	Temporary personnel and emergency teachers as well as faculty researchers below instructors are not included in the study.

Data Analysis Procedure

The data from the quantitative component is processed in the SPSS software in order to extract the percentage and frequency from the survey questionnaire. The open-ended interview was processed through a thematic analysis and evaluated alongside its quantitative counterpart. The results of these components then assessed by the researchers with the results of the primary data evaluation.

Figure 1. Triangulation Design: Validating Quantitative Data Model Framework



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(14) Major Activities/Work plan

Table 2. Activity workplan of the study.

	2023 Activity Workplan											
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Requesting of Permissions and Gathering of Preliminary Data												
Field Work – Gathering of Quantitative Data												
Analysis of Quantitative Data												
Field Work – Gathering of Qualitative Data												
Analysis of Quantitative Data												
Gathering of Primary Data												
Content Analysis												
Triangulation of Data and Synthesis of Research Paper												
Presentation												

Results and Discussions

Level of the Research Capability of the CAPSU Faculty

The level of the research capability of the respondents is expressed in table 3 with a grand mean of 3.67 verbally interpreted as “Highly Engaged.”

Table 3. Level of the research capability of the respondents.

Level of the Research Capability of the CAPSU Faculty	Mean	Verbal Interpretation
Technical Skills	3.85	Highly Engaged
Skills in Conceptualizing a Research Problem	3.76	Highly Engaged
Knowledge and Skills in Designing the Research Plan	3.63	Highly Engaged
Skills in Research Data Processing	3.43	Moderately Engaged
Grand Mean	3.67	Highly Engaged

It appeared that the level of the research capability of the respondent’s grand mean is 3.67, interpreted as “Highly Engaged.” This value reflects an overall moderate to high level of research capability among the CAPSU faculty, indicating a commendable proficiency across the different aspects of research. It signifies the presence of a solid foundation and competence among the faculty members, suggesting a conducive environment for research and scholarly activities within the institution.

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In relation to the implications and the cited references, Abrazado and Coronel (2016) may offer insights into the research capability and workload profile of teachers, providing valuable context for understanding the dynamics of research productivity within educational settings. Bowsher et al. (2019) may contribute perspectives on health research capacity strengthening in low and middle-income countries, providing valuable lessons applicable to various research contexts, including those in conflict-affected areas. The Capiz Quickstat from the Philippine Statistics Authority can provide additional context specific to the Capiz region, potentially offering valuable information on research capabilities and trends within the area.

These references collectively contribute to the understanding of the factors influencing research capability and productivity, providing valuable insights into the strategies and approaches that can be adopted to enhance research capacity and foster a culture of scholarly excellence within academic institutions.

Technical Skills

The level of the research capability of the respondents in terms of technical skills is presented in table 4.

Table 4. Level of the research capability of the respondents in terms of technical skill.

Technical Skills	Mean	Description
Written communication.	3.97	Highly Capable
Oral communication.	3.95	Highly Capable
Critical/analytical thinking.	3.90	Highly Capable
Problem solving.	3.83	Highly Capable
Use of computer commands/programs/software (ICT).	3.81	Highly Capable
Acknowledging or citing sources/cross-referencing.	3.81	Highly Capable
Online research, use of electronic resources databases and search engines (ICT).	3.79	Highly Capable
Research organization.	3.77	Highly Capable
Grand Mean	3.85	Highly Capable

The competence of the respondents technically for research is remarkable, given an overall grand mean of 3.85, which translates to “Highly Capable”. This implies that the participants show impressive skills in numerous technical fields for conducting research.

“Written Communication” with a mean score of 3.97 as “Highly Capable” and the highest mean suggests that the respondents are particularly good at communicating their thoughts and research findings through written media. This indicates that they understand the essence of the academic communication and knowledge dissemination which involves the ability to simplify the complex concepts and the facts.

On the other hand, the lowest mean of 3.77 is associated to Research Organization technical skill which is still interpreted as “Highly Capable”. This signifies a good capability but still there is

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an area for improvement regarding material and data organization which could result into improved research processes.

The conclusions of the study indicate that the respondents have a solid technical foundation necessary to conduct fruitful research. This aligns with the study of Vecaldo (2019) with the notion that a strong proficiency in technical skills is essential for conducting high-quality and impactful research in various fields. The findings underscore the significance of nurturing and further enhancing these technical competencies among researchers, which can be achieved through targeted training programs, workshops, and other educational initiatives. This not only enhances the quality of research output but also contributes to the overall advancement of knowledge and the academic community.

Skills in Conceptualizing a Research Problem

The level of the research capability of the respondents in terms of skills in conceptualizing a research problem is presented in table 5.

Table 5. Level of the research capability of the respondents in terms of skills in conceptualizing a research problem.

Skills in Conceptualizing a Research Problem	Mean	Description
Developing conceptual framework.	3.95	Highly Capable
Reviewing related literature.	3.81	Highly Capable
Formulating the research objectives or statement of the problems.	3.76	Highly Capable
Formulating a hypothesis or proposition.	3.74	Highly Capable
Defining concepts, constructs, or variables.	3.72	Highly Capable
Identification of a research problem or topic.	3.71	Highly Capable
Developing theoretical framework.	3.66	Highly Capable
Grand Mean	3.76	Highly Capable

In general, the respondents' research capability in relation to their ability to conceptualize a research problem has an overall grand mean of 3.76, which translates to "Highly Capable." Such a fact reveals the high performance of the participants in different essential dimensions of conceptualizing research problems.

The highest mean skill was "Developing a conceptual framework". Its mean was 3.95, which suggested that the respondents could create a good theoretical framework, guiding them through the research process. This means that they are able to develop a well-defined structure to support them in formulating and carrying out the experiment.

However, the lowest mean skill, "Developing a theoretical framework," has a mean of 3.66, which is still considered "Highly Capable". This implies that while the study displays some significant level of competences on this matter, there is still more to be improved by This might make the research more theoretically sound among the respondents.

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In the context of the study, the results highlight the importance of a sound conceptualization of research problems, emphasizing the critical role of skills such as developing conceptual frameworks and reviewing related literature. These findings are supported by Kurt (2018) in their exploration of the factors motivating authors to publish in predatory journals, underlining the significance of a strong conceptual foundation in research to ensure the credibility and integrity of scholarly work.

Knowledge and Skills in Designing the Research Plan

The level of the research capability of the respondents in terms of knowledge and skills in designing the research plan is presented in table 6.

Table 6. Level of the research capability of the respondents in terms of knowledge and skills in designing the research plan.

Knowledge and Skills in Designing the Research Plan	Mean	Description
Ethical consideration	4.00	Highly Capable
Study locale	3.82	Highly Capable
Sources of data	3.78	Highly Capable
Research design	3.76	Highly Capable
Population and sampling techniques	3.43	Moderately Capable
Types of Instruments	3.42	Moderately Capable
Procedures in data collection.	3.40	Moderately Capable
Statistical Tools	3.40	Moderately Capable
Grand Mean	3.63	Highly Capable

It appeared the research capability of the respondents concerning knowledge and skills in designing the research plan with a grand mean of 3.63, interpreted as “Highly Capable.” This suggests that the participants possess a strong proficiency in various aspects related to designing a comprehensive research plan.

The knowledge and skill with the highest mean is “Ethical Consideration,” with a mean of 4.00, indicating that the respondents demonstrate an exceptional understanding of ethical standards and considerations in research. This finding highlights the respondents’ meticulous approach to upholding ethical integrity and ensuring the protection of human subjects and the integrity of their research outcomes.

Conversely, the knowledge and skills with the lowest means are “statistical tools,” “with a mean of 3.40. Although these scores suggest a satisfactory level of capability, they also indicate a potential area for improvement in the understanding and application of appropriate statistical tools to produce meaningful and of appropriate results.

In the context of the study, the findings underscore the critical role of ethical considerations in research planning, aligning with the perspectives of Chu et al. (2014) on the importance of fostering equity and global health collaborations through a strong emphasis on ethical research practices. Additionally, the insights presented are in line with the recommendations by Huenneke

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et al. (2017) on key strategies for enhancing the research capacity of university faculty members, emphasizing the significance of a robust understanding of ethical considerations in research planning.

Furthermore, the references by Creswell (2014) and Kotsemir & Shashnov (2017) emphasize the importance of a comprehensive understanding of various research methodologies and tools, supporting the need for a more nuanced grasp of sampling techniques, research instruments and statistical tools to ensure the overall quality and validity of research outcomes.

Skills in Research Data Processing

The level of the research capability of the respondents in terms of skills in research data processing is presented in table 7.

Table 7. Level of the research capability of the respondents in terms of skills in research data processing.

Skills in Research Data Processing	Mean	Description
Analysis of data	3.46	Highly Capable
Interpretation of data	3.45	Highly Capable
Statistical Treatment data for quantitative research	3.42	Moderately Capable
Presentation of data (textual, graphical, tabular)	3.42	Moderately Capable
Treatment data for qualitative research (coding)	3.41	Moderately Capable
Grand Mean	3.43	Moderately Capable

It appeared that the research capability of the respondents in terms of skills in research data processing is moderately capable, with an overall grand mean of 3.43. The findings indicate that the participants possess a solid foundation in research data processing, but there is still room for improvement in certain areas to enhance the overall quality and comprehensiveness of their data processing abilities.

The skill with the highest mean is “Analysis of data,” with a mean of 3.46, indicating that the respondents demonstrate a high level of competence in analyzing research data. This suggests their proficiency in applying appropriate analytical techniques to interpret and draw meaningful insights from the data, contributing to the depth and richness of their research findings.

Conversely, the skills with the lowest means are “Treatment data for qualitative research (coding),” with a moderately capable mean of 3.41. These scores suggest that while the respondents exhibit a satisfactory level of competency in this area, further refinement in their understanding and application of qualitative research and data presentation techniques could strengthen the comprehensiveness and impact of their research outputs.

It emphasizes the critical role of data processing skills in research, aligning with the observations made by Bowsher et al. (2019) on health research capacity strengthening in low and middle-income countries. These findings underscore the importance of enhancing data processing

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capabilities to foster more robust and impactful research in various contexts, including conflict-affected areas.

Furthermore, the insights presented are consistent with the Capiz Quickstat (2018) provided by the Philippine Statistics Authority, emphasizing the need for a comprehensive approach to data processing and analysis to ensure the accuracy and reliability of statistical information. It underscores the significance of continuous development in research data processing skills, highlighting the importance of adopting effective analytical techniques, enhancing data presentation methodologies, and fostering a more nuanced understanding of statistical treatment to facilitate the generation of valuable and reliable research insights.

Level of Extent of Participation of Other Stakeholders

The level of extent of participation of other stakeholders expressed in table 8 with a grand mean of 3.65 verbally interpreted as “Highly Engaged”.

Table 8. Level of teaching competency of the respondents.

Level of Extent of Participation of Other Stakeholders	Mean	Verbal Interpretation
Stakeholders Engagement	3.69	Highly Engaged
Collaboration and Partnership	3.62	Highly Engaged
Grand Mean	3.65	Highly Engaged

The grand mean, as calculated from the data in Table 1, is 3.65, and it is interpreted as “Highly Engaged.” This value signifies that the respondents exhibit a strong level of commitment and active involvement in stakeholder engagement, collaboration, and partnership activities. The demonstrated engagement suggests a positive and productive relationship between the respondents and the various stakeholders, highlighting a culture of cooperation and synergy within the academic community.

In the context of the implications, the references provided can support the notion of the importance of active engagement and collaboration within academic and research environments. These references may contribute to an understanding of the significance of fostering robust relationships and partnerships, ultimately enhancing the research capability and productivity of academic institutions, particularly in low and middle-income countries. The references may offer valuable insights into the strategies and practices that can be implemented to strengthen research capacity and foster effective collaboration, promoting growth and development within educational and research communities.

Abrazado and Coronel (2016) may provide insights into the research capability and workload profiles of educators, emphasizing the significance of productive engagement and collaboration in enhancing research productivity. Bowsher et al. (2019) may offer perspectives on health research capacity strengthening, underlining the importance of collaboration, particularly in conflict-affected areas. The Capiz Quickstat from the Philippine Statistics Authority may provide

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contextual information relevant to the region, offering insights into the significance of collaborative initiatives for regional development and progress.

Stakeholders Engagement

The level of extent of participation of the respondents to other participants of the respondents in terms of stakeholders' engagement is presented in table 9.

Table 9. Level of extent of participation of the respondents to other participants of the respondents in terms of stakeholders' engagement.

As a faculty researcher, I...	Mean	Description
10. Believe the involvement has a positive impact on the innovation ecosystem.	3.81	Highly Engaged
7. Capiz State University appreciates the feeling of acceptance by valuing their opinions and ideas.	3.75	Highly Engaged
1. Engage in the innovation endeavors of Capiz State University.	3.74	Highly Engaged
8. View the university as a dependable collaborator in cultivating advancement.	3.72	Highly Engaged
3. Work together efficiently with the university to execute collaborative ventures.	3.71	Highly Engaged
2. Offer substantial insights and expertise to improve the innovation ecosystem.	3.68	Highly Engaged
9. Satisfied with the level of communication and interaction with the university.	3.68	Highly Engaged
6. Provide comprehensive information regarding the innovation goals and priorities of the university.	3.67	Highly Engaged
5. Partake in activities to exchange knowledge and enhance skills.	3.60	Highly Engaged
4. Allocate financial funds to back innovation projects.	3.53	Highly Engaged
Grand Mean	3.69	Highly Engaged

The respondents have a high level of engagement with various stakeholders in the context of their role as faculty researchers. The overall grand mean for the extent of participation in stakeholders' engagement is 3.69, which is interpreted as "Highly Engaged." This suggests that the respondents actively and positively contribute to fostering a collaborative and innovative ecosystem within Capiz State University.

It appeared that the highest mean score of 3.81 was attributed to the statement, "Believe the involvement has a positive impact on the innovation ecosystem," highlighting the perceived positive influence of active engagement on the overall innovation environment. This positive perception among the faculty researchers underscores the importance of their involvement and the acknowledgment of their contribution to the university's innovation landscape.

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It of Bowsher, et. Al. (2016), emphasizing the critical role of stakeholder engagement in fostering a culture of innovation within academic institutions. Their findings highlight the significance of active collaboration, communication, and financial support from stakeholders in driving successful innovation initiatives within the university setting. The implications of these findings underscore the importance of fostering a culture of active participation and collaboration among stakeholders, including faculty researchers and other participants. Additionally, it emphasizes the need for a more comprehensive approach to resource allocation, particularly in terms of financial support, to bolster innovation projects within the university. Creating a conducive environment for stakeholders to contribute and share their expertise actively can significantly enhance the overall innovation ecosystem and contribute to the university's long-term growth and development.

Collaboration and Partnership

The level of extent of participation of the respondents to other participants of the respondents in terms of collaboration and partnership is presented in table 10.

Table 10. Level of extent of participation of the respondents to other participants in terms of collaboration and partnership.

As a faculty researcher, I...	Mean	Description
2. Exchange of ideas and expertise between the university and external stakeholders, fostering a culture of collaboration.	3.71	Highly Engaged
1. Establish formal collaborative agreements with industry partners to promote innovation and knowledge transfer.	3.70	Highly Engaged
4. Seek and value partnerships with government agencies to leverage resources and support for innovation activities.	3.65	Highly Engaged
7. Maintain regular communication channels with local communities to identify opportunities for collaboration and address their needs.	3.63	Highly Engaged
6. Cooperate with research institutions and other universities has facilitated joint research projects and publications.	3.61	Highly Engaged
8. Manage and nurture partnerships with external stakeholders.	3.60	Highly Engaged
9. Establish mechanisms for evaluating the effectiveness and impact of collaborative initiatives between the university and external partners.	3.58	Highly Engaged
3. Collaborate projects and initiatives between the university and external partners have resulted in mutual benefits and positive outcomes.	3.57	Highly Engaged
10. Provide incentives or recognition to faculty and staff for active activities and partnerships.	3.57	Highly Engaged
5. Create a structured framework for engaging in innovation-related projects and mentorship opportunities.	3.54	Highly Engaged

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Grand Mean

3.62 Highly Engaged

It appeared that the level of extent of participation of the respondents to other participants in terms of collaboration and partnership with the overall grand mean for the extent of participation in collaboration and partnership activities is 3.62, which is interpreted as “Highly Engaged.” This suggests that the respondents are actively involved in establishing and nurturing collaborative relationships with external entities, thereby contributing to the promotion of a culture of cooperation and innovation within Capiz State University.

Among the statements evaluated, the highest mean score of 3.71 was attributed to the statement, “Exchange of ideas and expertise between the university and external stakeholders, fostering a culture of collaboration.” This research highlights the value of ongoing interaction and information sharing between the institution and outside partners, pointing to a strong collaborative culture that promotes creative thinking and the sharing of knowledge.

The statement "Create a structured framework for engaging in innovation-related projects and mentorship opportunities," on the other hand, received the lowest mean score of 3.54. This finding indicates that there is room for improvement in the creation of a comprehensive framework to support and expedite participation in innovation-related projects and mentorship opportunities. Enhancing this framework further might lead to a more organized approach to innovation nurturing and further boost the university's joint efforts.

The findings are consistent with the study carried out by Huenneke et al. (2017), which highlights the vital role that cooperative partnerships have in promoting innovation and accelerating the transfer of information between academic institutions and outside stakeholders. Their findings highlight the value of establishing an atmosphere that is favorable to the sharing of information, resources, and skills and highlight the necessity of formal frameworks in order to facilitate and maintain cooperative projects. These results highlight how crucial it is to keep working to build and maintain relationships with a range of external stakeholders, such as business associates, governmental organizations, regional communities, and academic institutions. Furthermore, the findings highlight the necessity of an all-encompassing framework that makes it easier to assess the outcomes and efficacy of cooperative projects, allowing the institution to better utilize its cooperative efforts and guarantee reciprocal gains.

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Descriptive Analysis of Demographics, Pertinent Papers, and other Parameters Socio-demographic Profile of the Respondents

The socio-demographic profile of the respondents in terms of age, sex, civil status, faculty rank, and highest educational attainment is presented in table 11.

Table 11. The sociodemographic profile of the respondents.

Socio-demographic Profile	Frequency	Percentage
Age		
21 – 35 Y/O	81	38
36-54 Y/O	88	41
52-65 Y/O	46	21
Total	215	100
Sex		
Male	78	36
Female	137	64
Total	215	100
Civil Status		
Single	99	46
Married	66	31
Widowed/Separated	50	23
Total	215	100
Faculty Rank		
Instructor	70	33
Assistant Professor	66	31
Associate Professor	74	34
Professor	5	2
Total	215	100
Highest Educational Attainment		
Bachelor Degree Holder	0	0
Master's Degree Holder	140	65
Doctorate Degree Holder	75	35
Total	215	100

Age. Out of 214 respondents, the highest frequency is observed in the age group of 36-54 years, which accounts for 88 (41.10%), followed by the age group of 21-35 years, which constitutes 81 (38.0%). The age group of 52-65 years has the lowest frequency of respondents, making up only 46 (21.0%).

Sex. Of the 215 respondents, the majority of the respondents were female 137 (64.0%), while male respondents accounted for 78 (36.0%).

Civil Status. Out of 215 respondents, the majority of the respondents were single, accounting for 99 (46.0%) of the sample, followed by married respondents 66 (31.0%) and widowed/separated respondents 50 (23.0%).

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Faculty Rank. Of the 215 respondents, the majority of the respondents were Associate Professor 74 (34.0%), followed by Instructor 70 (33.0%), and Assistant Professor 66 (31.0%). A small percentage of the respondents were Professor 5 (2.0%)

Highest Educational Attainment. Out of the 215 respondents, the majority of the respondents 140 (65.0%) had a Master's Degree, while 75 (35.0%) of the respondents had a Doctorate Degree. No percentage of the respondents 0 (0%) had a Bachelor's Degree.

Pertinent Papers and Other Parameters of the Respondents

The pertinent papers and other parameters of the respondents in terms of training/seminars attended, research undertaking, number of research outputs presented, and number of research outputs presented are presented in the tables.

Table 12. Training/seminars attended by the respondents.

Training/Seminars Attended:	Frequency	Percentage
International	81	38
National	83	39
Regional	215	100
School	215	100

A significant proportion of respondents, accounting for 100%, participated in regional events, indicating a strong commitment to local collaboration and knowledge exchange within the immediate geographical sphere. Furthermore, the data indicates a noteworthy engagement in national (39%) and international (38%) activities, reflecting the university's active involvement in broader innovation networks beyond its regional context. This balanced participation across different levels suggests that CAPSU is strategically positioned to contribute to local and global innovation landscapes.

The balanced participation across different levels of training and seminars suggests that Capiz State University (CAPSU) is strategically positioned to contribute to local and global innovation landscapes. This finding aligns with the literature on building research capacity in university faculty members (Huenneke et al., 2017), emphasizing key strategies to enhance research capabilities. The active engagement in regional, national, and international events reflects the university's commitment to fostering collaboration and knowledge exchange, essential elements for a robust innovation ecosystem.

Table 13. The research undertaken by the respondents.

Research Undertaking	Frequency	Percentage
Completed	81	37%
Proposed	83	39%

It reveals a balanced distribution between completed and proposed research undertakings among the respondents. Notably, 37% of the research endeavors have been successfully completed, showcasing the university's commitment to advancing knowledge and contributing to the

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academic and innovation landscape. On the other hand, 39% of the research initiatives are currently in the proposed stage, indicating an ongoing and dynamic research agenda at CAPSU. The balanced distribution between completed and proposed research undertakings among respondents at Capiz State University (CAPSU) suggests a multifaceted research environment. This finding resonates with the work of Kotsemir and Shashnov (2017), who emphasize the importance of measuring, analyzing, and visualizing research capacity at the departmental and staff levels. The successful completion of 37% of research endeavors underscores CAPSU's effectiveness in translating research proposals into tangible outcomes. Simultaneously, the ongoing 39% of proposed research initiatives reflects the university's proactive stance in maintaining a robust and evolving research agenda.

Table 14. No. of research output presented by the respondents.

No. of Research Output Presented	Frequency	Percentage
Local	103	48
Regional	30	14
National	28	13
International	28	13

48% of the research has been presented locally, demonstrating CAPSU's dedication to meeting and even exceeding local demands. A larger influence within the university's geographic environment is also shown by the fact that 14% of research outcomes have reached the regional level. Research presentations at the national (13%) and international (13%) levels are distributed in a way that highlights CAPSU's involvement in larger academic and innovation networks and reflects its goal of keeping up with and contributing to developments outside of the local and regional domains. The results point to a thorough and well-rounded approach to knowledge transmission at CAPSU, focusing on addressing regional and global issues through compelling research presentations.

Capiz State University's (CAPSU) dedication to a thorough and well-rounded approach to information transmission is suggested by the varied distribution of research presentations at various levels (local, regional, national, and worldwide). This is consistent with the idea of international cooperation and participation in broader scholarly networks, as covered by Dutta et al. (2021) in the Global Innovation Index. The results demonstrate how committed CAPSU is to meeting local needs (48% at the local level) and how involved it is in larger academic and innovation networks at the regional, national, and worldwide levels (14%, 13%, and 13%, respectively).

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Table 15. No. of research output published by the respondents.

No. of Research Output Published	Frequency	Percentage
Local	43	20
Regional	9	4
National	7	3
International	20	9

20% of research results made locally have been published, demonstrating CAPSU's dedication to meeting local needs. In the region, just 4% of research has been published, indicating a small yet noteworthy influence within the university's geographic area. 3% and 9%, respectively, at the national and international levels show how involved CAPSU is in larger academic and innovation networks. National and international publications indicate a commitment to contributing to the larger academic discourse and keeping up with global innovations, while the emphasis on local and regional publications highlights the university's attention to tackling urgent concerns. The results suggest that CAPSU should take a multimodal strategy to disseminating research, aiming for both local and wider scholarly effect.

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Possible Technology/Software for Faculty Researchers in Conducting Studies

The thematic analysis of the statements of the participants on the possible technology/software for faculty researchers in conducting studies is shown in table 15.

Table 15. Possible technology/software for faculty researchers in conducting studies.

Code	Category	Theme(s) Extracted
QGIS	Graphical Information	Geospatial Data Visualization
R Programming, Jamovi, JASP	Statistical Analysis	Statistical Software

The statements from faculty researchers at Capiz State University provide valuable insights into the preferences and experiences regarding technology and software for conducting studies. Notably, participants demonstrated a distinct preference for specific tools, with QGIS emerging as a standout choice for geospatial data visualization. The participants commended QGIS for its user-friendly interface and effective graphical representation of data, suggesting its suitability for geospatial research projects.

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One participant FG112 said, "*Mas gusto ko gamiton ang QGIS sa akon geospatial nga study. Madali siya gamiton kag maayo ang graphical representation sang data,*" (FGP118) reflecting the positive sentiment towards QGIS's ease of use and graphical capabilities.

Furthermore, R Programming garnered recognition for its statistical analysis capabilities, despite being acknowledged as somewhat complex. The acknowledgment of R Programming's power implies a willingness among faculty members to engage with advanced statistical tools, highlighting the importance of such tools in their research endeavors.

Participant FG002 stated, "*Mas Prefer ko ang R Programming sa akon statistical analysis kaso medyo komplikado pero powerful siya,*" (FGP116) underscoring the perceived strength of R Programming in statistical analysis.

Jamovi emerged as a preferred statistical software, attributed to its user-friendly nature and the immediate delivery of statistical insights. The absence of a specific statement for JASP suggests potential interest or consideration for statistical analysis, though more explicit insights from participants would be beneficial.

Drawn from these preferences is the need for a diverse set of tools to cater to different preferences and project requirements among faculty researchers. Capiz State University should consider investing in training programs to enhance faculty proficiency in these identified software/tools. Fostering a collaborative environment for knowledge sharing would facilitate the exchange of experiences and insights. Additionally, staying abreast of emerging technologies is crucial to ensuring the continuous improvement of research capabilities within the institution.

"Jamovi gid ang akon ginagamit, kay user-friendly siya kag makahatag dayon sang mga statistical insights," (FGP110).

It emphasizes the user-friendly nature of Jamovi and its immediate contribution to statistical insights, further supporting its favorable position among faculty researchers. Overall, the identified technologies, including QGIS, R Programming, Jamovi, and JASP, represent a promising array of tools for faculty researchers at Capiz State University to conduct diverse and sophisticated studies.

Training Needs Programs of the Faculty Researchers in Conducting Studies

The thematic analysis of the statements of the participants on the training needs programs of the faculty researchers in conducting studies is shown in table 16.

Table 16. Training needs programs for the faculty researchers in conducting studies.

Code	Category	Theme(s) Extracted
Infographics, motion graphics	Data Visualization	Visual Representation
Data-driven decision-making, statistical methods	Statistical Analysis	Statistical Tools

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The statements from faculty researchers at Capiz State University illuminate critical insights into the training needs programs essential for enhancing their capabilities in conducting studies. The identified themes primarily revolve around data visualization and statistical analysis. Under the category of data visualization, participants express a consistent challenge in selecting appropriate statistical methods and emphasize the necessity for training in visual representation techniques, such as infographics and motion graphics.

“Permi ko nagakabudlay sa pagpili sang husto nga statistical method para sa akon data analysis kaangay sa infographics kag motion graphics. Permi ko mabatyagan nga nagalumog ako sa damu nga data, kag mabudlay gid ma-extract sang data kis-a, (FGP009).

The participants articulate the struggle of feeling inundated by voluminous data, making data extraction a complex task. In response to the challenges of statistical analysis, particularly in data-driven decision-making, there is a clear demand for training programs focusing on statistical tools and methods. The need for assistance in plotting motion graphics for presenting statistics further underscores the challenges faced by researchers.

“Damo gid nga mga manug-research ang nagakabudlay sa sini. Sa pagplastar sang mga motion graphics sa pagpresent kg statistics sang mga data, (FGP029).

The overarching implication is the imperative development of tailored training programs aimed at improving researchers' proficiency in selecting appropriate statistical methods and enhancing their skills in visually representing data through infographics and motion graphics. These training initiatives should foster a supportive learning environment, addressing the challenges associated with the abundance of data and promoting interdisciplinary approaches that integrate statistical methods with effective data-driven decision-making strategies. In essence, the identified training needs programs highlight an opportunity for targeted interventions to elevate the research capabilities of faculty members at Capiz State University.

Problems/Issues Encountered in Making and Conducting a Research Output

The problems/issues in making and conducting a research output is shown in table 17.

Table 17. The respondents' problems/issues in making and conducting a research output.

Problems	Frequency	Percentage
Lack of time	210	98
Financial Problems	200	93
Pre-occupied with teaching loads and paperwork.	200	93
Lack of the necessary skills	89	41
Lack of training/workshops in research	50	23

An overview of the challenges faced by respondents in the process of creating and conducting research output at Capiz State University. The most prevalent issue, highlighted by a substantial 98% of respondents, is the constraint of time. This underscores the significant time demands on faculty members and researchers, which may impede their ability to dedicate ample time to research endeavors. Financial problems also emerge as a prominent challenge, with 93% of

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respondents citing financial constraints as a hindrance to their research initiatives. Additionally, an equal percentage of respondents (93%) express struggles arising from the dual burden of teaching loads and paperwork, emphasizing the competing demands on faculty members' time and energy. Furthermore, 41% of respondents identify a lack of necessary skills as a barrier, underscoring the importance of ongoing professional development. Lastly, 23% of respondents note the absence of training and workshops in research, suggesting a potential gap in capacity-building opportunities. Addressing these challenges holistically, such as through targeted training programs and resource allocation, could enhance the research ecosystem at Capiz State University.

(16) Conclusions and Recommendations

The research demonstrates that Capiz State University's faculty possesses a commendable and highly engaged level of research capability across various domains, indicating a strong foundation for scholarly activities. It demonstrates a highly engaged level of participation in stakeholder engagement, collaboration, and partnership activities, fostering a positive and productive relationship within the academic community and beyond. The descriptive analysis of demographics, pertinent papers, and other parameters reveals a diverse and actively engaged faculty at Capiz State University, with balanced participation in training/seminars, a dynamic mix of completed and proposed research initiatives, and a comprehensive approach to disseminating research outputs at local, regional, national, and international levels. The possible technology/software for faculty researchers in conducting studies are QGIS in geospatial data visualization and recognition of the power of R Programming for statistical analysis, emphasizing the importance of providing a diverse set of tools to cater to various preferences and project requirements. It has critical need for tailored training programs in data visualization, specifically focusing on the challenges of selecting appropriate statistical methods and improving skills in visual representation techniques like infographics and motion graphics, illuminating a key opportunity for enhancing the research capabilities of faculty researchers at Capiz State University. The challenges identified in this research output assessment at Capiz State University, particularly the overwhelming constraints of time, financial limitations, teaching loads, and skill gaps, collectively emphasize the multifaceted obstacles faced by faculty researchers, highlighting the need for comprehensive strategies to address these issues.

The recommendation is to further enhance research prowess, targeted training programs should prioritize refining skills in statistical treatment for quantitative research, improving data presentation, and fostering continuous development in theoretical frameworks and organizational aspects of research. To strengthen the university's innovation ecosystem, it is recommended to focus on enhancing the financial support framework for innovation projects, creating a more robust structure for engaging in innovation-related projects and mentorship opportunities, and continually evaluating the effectiveness and impact of collaborative initiatives to ensure sustained mutual benefits for all stakeholders involved. It must continue fostering a supportive environment for diverse training opportunities, encourage the progression of proposed research initiatives into completion, and further promote the dissemination of research outputs, especially at the regional and national levels, to amplify the university's impact on both local and broader academic communities. To enhance research capabilities, Capiz State University is recommended to invest in training programs that focus on QGIS, R Programming,

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Jamovi, and JASP, fostering a collaborative environment for knowledge sharing among faculty members and staying updated on emerging technologies to improve research proficiency within the institution continually. It is also recommended to design and implement targeted training programs addressing the identified needs in data visualization, statistical analysis, and visual representation, creating a supportive learning environment to empower faculty researchers in overcoming challenges associated with data abundance and fostering interdisciplinary approaches to strengthen their research proficiency. It is recommended to implement a holistic approach, including time-management support, financial resources allocation, and targeted training programs to enhance faculty members' skills, mitigating the challenges associated with time constraints, financial limitations, teaching responsibilities, and skill gaps, thereby fostering a more conducive environment for impactful research endeavors.

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